

# Removal of the spatial cutoff for weakly coupled $P(\varphi)_2$ : norm convergence of the ground-state net on the quasi-local algebra (M6, v2 — audited repair)

## Abstract

Let  $H(g)$  be the spatially cutoff  $P(\varphi)_2$  Hamiltonian,  $\Omega_g$  its ground state, and  $\omega_g$  the induced state on the time-zero quasi-local algebra  $\mathfrak{A}$ . Glimm and Jaffe (1970) constructed an infinite-volume vacuum as a  $w^*$  *limit point* of space-averaged  $\omega_g$ ; the cited construction does not prove convergence of the net itself. Guerra–Rosen–Simon (1975) formulated analogous Euclidean local-convergence questions as their Problems 1–2. We prove: in the Glimm–Jaffe–Spencer weak-coupling region  $0 \leq \lambda/m_0^2 < \varepsilon_{\mathcal{P}}$ , the *full net*  $(\omega_g)$  — no subnets, no averaging — converges in norm on every local algebra,

$$\lim_{g \in \mathcal{G}} \|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| = 0,$$

with an explicit super-polynomial rate in  $\text{dist}(O, \{g < 1\})$ ; the limit is locally normal, translation invariant,  $\mathbb{Z}_2$ -invariant for even  $\mathcal{P}$ , and is a ground state of the Glimm–Jaffe dynamics  $(\mathfrak{A}, \alpha)$  in the spectral sense. The proof is Hamiltonian: a local-perturbations-perturb-locally estimate built on an exact spectral filter and the exact light cone makes the net Cauchy; its two hypotheses — a cutoff-uniform gap  $\text{spec } H(g) \subseteq \{0\} \cup [m_1, \infty)$  and a uniform local interaction moment — are derived from the two printed cluster-expansion theorems of Glimm–Jaffe–Spencer (Ann. of Math. 100 (1974), Thms. 1.1.7 and 1.1.8), whose cutoff-uniformity is both printed (p. 629) and audited. The proof also imports, and explicitly identifies below, the standard cutoff-construction results on self-adjointness and the simple positive ground state, Feynman–Kac–Nelson transfer, finite-volume stability and hypercontractivity, domain smoothing and a common core, finite propagation and patching, and covariance. No Euclidean uniqueness theorem, global Markov property for the infinite-volume measure, or reverse Osterwalder–Schrader reconstruction is used. We do not claim uniqueness of  $\omega_\infty$  among all locally normal ground states, a sharp exponential rate, or anything outside the weak-coupling region.

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# 1 Introduction

## 1.1 The problem

The  $\lambda\varphi_2^4$  (more generally  $P(\varphi)_2$ ) model was constructed by Glimm and Jaffe in the series I–IV [1, 2, 3, 4]. Paper III builds the physical vacuum as follows: the finite-volume ground states  $\Omega_g$  of the cutoff Hamiltonians  $H(g)$  induce states  $\omega_g$  on the quasi-local algebra  $\mathfrak{A}$ ; after a space-translation averaging, a  $w^*$ -compactness argument produces a *limit point*  $\omega$ , which is shown to be locally Fock [3, Thms. 2.1–2.3]. Whether the net  $(\omega_g)$  *converges* — so that the vacuum is canonical rather than selected by a limit procedure — was not proved there; the Expositions volume states plainly that the  $g \rightarrow 1$  limit of the vacuum expectation values “has not been proved” [6, §4]. Guerra, Rosen and Simon list analogous *Euclidean* local  $L^p$  and finite-volume Gibbs-state convergence statements as Problems 1–2 of [10, pp. 125–126]; their note added on p. 128 records Newman’s solution of the small-coupling  $p = 1$  versions. (Some later surveys state the Hamiltonian cutoff limit as a convergence; the cited primary construction itself is unambiguously a limit-point construction.)

On the Euclidean side, convergence of *Schwinger functions* at weak coupling follows from the cluster expansion of Glimm–Jaffe–Spencer [7]. Albeverio–Høegh-Krohn–Zegarlinski prove Euclidean Gibbs uniqueness under their stated hypotheses [13]; Bauerschmidt–Dagallier–Weber prove uniqueness of the invariant measure for the  $\Phi_2^4$  stochastic-quantization equation above the critical temperature [14]. Neither statement is, by itself, a theorem about norm convergence of this Hamiltonian ground-state cutoff net. The cited primary construction does not prove that local-norm convergence, and no such theorem is identified in the references reviewed here. The Hamiltonian-renormalisation programme of Rodriguez Zarate–Thiemann cited here is explicitly restricted to finite volume [15].

## 1.2 The result

*In the GJS weak-coupling region, the full net  $(\omega_g)_{g \in \mathcal{G}}$  converges in norm on every local algebra; the limit is locally normal, translation and (for even  $\mathcal{P}$ )  $\mathbb{Z}_2$  invariant, and is a ground state of the Glimm–Jaffe dynamics. An explicit rate holds:  $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \Psi_{m_1}(d)$  for  $g \equiv 1$  on  $O_d$ , with  $\Psi_{m_1}$  decaying faster than every inverse power.*

See Theorem 7.2. The logical structure is deliberately Hamiltonian and compactness-free:

| step  | content                                                                      | where proved   |
|-------|------------------------------------------------------------------------------|----------------|
| (i)   | printed GJS cluster/moment theorems, cutoff-uniform                          | [7] + audit §3 |
| (ii)  | uniform local moment (M <sub>2</sub> )                                       | §4             |
| (iii) | uniform gap (G): $\text{spec } H(g) \subseteq \{0\} \cup [m_1, \infty)$      | §5             |
| (iv)  | LPPL/Cauchy: (G)+(M <sub>2</sub> ) $\Rightarrow$ full-net convergence + rate | §6             |
| (v)   | the limit is an $\alpha$ -ground state; invariances                          | §7             |

Steps (ii)–(iii) convert Euclidean input to Hamiltonian statements through slab Feynman–Kac–Nelson transfers in which every limit is taken at fixed  $g$ ; no uniform gap is presupposed anywhere in their proofs (the gap of step (iii) is an output). Step (iv) replaces the compactness of [3] by a Cauchy argument: the same gap that yields clustering also yields a local-perturbations-perturb-locally estimate, which bounds  $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\|$  whenever both cutoffs equal 1 on  $O_d$ . Step (v) is the exact spectral identity of the companion note M0: the ground-state property of the limit costs an identity, not an estimate.

### 1.3 What is not claimed

(a) *Uniqueness*: Theorem 7.2 identifies the limit of *this* cutoff family; it does not show that every locally normal ground state of  $(\mathfrak{A}, \alpha)$  equals  $\omega_\infty$ . (b) *Sharp rate*: in the exactly solvable quadratic model the true rate is  $e^{-2m_1 d}$  (companion note M2);  $\Psi_{m_1}$  is super-polynomial but not exponential, and this is intrinsic to the filter method used. (c) *Beyond weak coupling*: under the explicit phase-classification hypotheses stated in Corollary 7.4, a cutoff-uniform gap is incompatible with that two-phase structure. Thus an extension to such a broken phase would have to abandon the uniform-gap input; no claim is made outside  $0 \leq \lambda/m_0^2 < \varepsilon_{\mathcal{D}}$ . (d) *Wavelet uniformity*: uniformity in a lattice/wavelet resolution is a separate theorem (Part II) and is not addressed here.

### Provenance

This is the v2 audited repair: all components are stated and proved in this document. The LPPL machinery of §6 originates in the companion files `m4/lppl.tex` and `m4/remaining.tex` and has been re-verified line by line during the merge (two sign/constant slips in the v0 pointers were corrected: the filter transform is  $+E^{-1}$ , the one-path bound carries no factor  $\frac{1}{2}$ ). This version additionally repairs the source pinpoints, the dimensionless scaling reduction, signed and unbounded FKN passages, the unit-cell count, the negative-time norm ratio, the exponential-moment prefactor, the spectral-projection range, and the hypotheses of the symmetry-breaking corollary. External inputs are the printed theorems of [7] (with the uniformity audit summarised in Remark 3.3) and the explicitly cited construction results from [12], [10], and [6]; companion notes are used only for provenance and extended discussion. Gate R2 (the norm prefactor of [7, (1.1.7)–(1.1.8)]) is closed by direct inspection of the page images:  $K_1$  occurs once and the local weight is  $K_2^{2N(\Delta)}$ .

## 2 Setting and standing assumptions

### 2.1 The model

$\mathcal{H} = L^2(\mathcal{S}'(\mathbb{R}), d\phi_0)$  is  $Q$ -space over the time-zero free field of mass  $m_0 > 0$ :  $d\phi_0$  is Gaussian with covariance  $C = (2\mu)^{-1}$ ,  $\mu = (-\partial_x^2 + m_0^2)^{1/2}$ . Weyl operators  $W(f, h) = \exp i(\varphi_0(f) + \pi_0(h))$  generate the local net

$$\mathfrak{A}(O) = \{W(f, h) : f, h \in C_c^\infty(O) \text{ real}\}'' , \quad \mathfrak{A} = \overline{\bigcup_O \mathfrak{A}(O)}^{\|\cdot\|}.$$

Let  $\mathcal{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$ ,  $a_{2p'} > 0$ , and for real  $h \in C_c^\infty$

$$V(h) := \lambda \int h(x) : \mathcal{P}(\varphi_0(x)) :_C dx,$$

a self-adjoint multiplication operator in  $Q$ -space, affiliated with  $\mathfrak{A}(U)$  for  $U \supseteq \text{supp } h$ , finitely additive on common domains ( $V(\sum h_j)\xi = \sum V(h_j)\xi$ ).

$$\mathcal{G} := \{g \in C_c^\infty(\mathbb{R}) : 0 \leq g \leq 1, g \equiv 1 \text{ on a nonempty bounded interval}\},$$

with  $g \preccurlyeq g'$  exactly when  $\{g = 1\}^\circ \subseteq \{g' = 1\}^\circ$ . For a bounded interval  $O$  and  $r \geq 0$ , set

$$O_0 := O, \quad O_r := \{x \in \mathbb{R} : \text{dist}(x, O) < r\} \quad (r > 0), \quad d(g, O) := \text{dist}(O, \mathbb{R} \setminus \{g = 1\}^\circ).$$

$$K(g) := H_0 + V(g), \quad E(g) := \inf \text{spec } K(g), \quad H(g) := K(g) - E(g),$$

with ground state  $\Omega_g$ ,  $\omega_g := \langle \Omega_g, \cdot \Omega_g \rangle$ ,  $\beta_t^g = \text{Ad } e^{itH(g)}$ .

## 2.2 Standing assumption (weak coupling)

Fix source-admissible constants  $M_* > 0$  and  $\lambda_* > 0$  such that the Glimm–Jaffe–Spencer expansion is valid at bare mass  $M_*$  whenever  $0 \leq \lambda' < \lambda_*$ . Define

$$\varepsilon_{\mathcal{P}} := \frac{\lambda_*}{M_*^2}.$$

Throughout,  $0 \leq \lambda/m_0^2 < \varepsilon_{\mathcal{P}}$ . After transport back to the original coordinates,  $m_1 > 0$  and  $C_7, C_8, K_1, K_2$  denote the constants in GJS Theorems 1.1.7–1.1.8 and their observable norms; only their positivity or finiteness and their cutoff independence will be used.

**Proposition 2.1** (dimensionless weak-coupling reduction). *The preceding inequality implies the source smallness hypothesis. More precisely, the GJS unitary dilation with  $s = m_0/M_*$  sends*

$$(m_0, \lambda) \longmapsto (s^{-1}m_0, s^{-2}\lambda) = \left(M_*, \frac{M_*^2}{m_0^2}\lambda\right),$$

and the transformed coupling is strictly smaller than  $\lambda_*$ .

*Proof.* For  $m > 0$ , write the Euclidean covariance, as an identity of tempered distributions, as

$$C_m(z) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \frac{e^{ip \cdot z}}{p^2 + m^2} dp.$$

Since  $sM_* = m_0$ , the substitution  $q = sp$  gives, without an omitted power of  $s$ ,

$$\begin{aligned} C_{M_*}(sz) &= \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \frac{e^{ip \cdot sz}}{p^2 + M_*^2} dp \\ &= \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \frac{s^{-2}e^{iq \cdot z}}{s^{-2}q^2 + M_*^2} dq = C_{m_0}(z). \end{aligned}$$

Thus the centered generalized Gaussian fields  $\Phi_{m_0}(x)$  and  $\Phi_{M_*}(sx)$  have identical joint covariances after smearing. This also fixes Wick ordering without a hidden constant. Indeed, for every regularized centered Gaussian variable  $X$ ,

$$:e^{tX}: = \exp\left(tX - \frac{1}{2}t^2\mathbb{E}[X^2]\right),$$

and the coefficient of  $t^n$  is  $:X^n:/n!$ . Apply the identity to a common mollifier, transported by  $x \mapsto sx$ , and then take the defining distributional limit. Equality of the regularized covariances shows that the unitary Gaussian identification sends  $:\mathcal{P}(\Phi_{m_0}(x)):_ {m_0}$  to  $:\mathcal{P}(\Phi_{M_*}(sx)):_ {M_*}$  with the coefficients of  $\mathcal{P}$  unchanged. For a Euclidean cutoff  $h$  the change of variables  $y = sx$  therefore gives

$$\lambda \int_{\mathbb{R}^2} h(x) : \mathcal{P}(\Phi_{m_0}(x)) :_{m_0} dx = s^{-2} \lambda \int_{\mathbb{R}^2} h(y/s) : \mathcal{P}(\Phi_{M_*}(y)) :_{M_*} dy.$$

The transformed cutoff  $h_s(y) := h(y/s)$  still satisfies  $0 \leq h_s \leq 1$  and has compact support. Hence the transformed parameters are exactly those displayed in the proposition, as stated in [7, p. 590]. Finally,

$$s^{-2} \lambda = \frac{M_*^2}{m_0^2} \lambda < M_*^2 \varepsilon_{\mathcal{P}} = \lambda_*.$$

This is precisely the source smallness condition. If the source decay rate in the scaled coordinate  $y = sx$  is  $m_{1,*} > 0$ , then  $e^{-m_{1,*} dy} = e^{-sm_{1,*} dx}$ ; thus the rate in the original coordinates is  $m_1 := sm_{1,*} > 0$ . The transported finite prefactors are the constants named above.  $\square$

### 2.3 Facts of the cutoff construction

**Source Fact 2.2.**  $K(g)$  is essentially self-adjoint and bounded below, and  $E(g)$  is a *simple* eigenvalue, for every measurable  $0 \leq g \leq 1$  of compact support;  $\Omega_g$  may be chosen  $> 0$  a.e. in  $Q$ -space. [7, p. 588]; [1, 2, 11, 5]

**Source Fact 2.3** (dynamics; the side conditions (P), (T)). There is a one-parameter automorphism group  $\alpha$  of  $\mathfrak{A}$ , independent of  $g$ , with

$$\alpha_t(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|}), \quad \alpha_t(A) = \beta_t^g(A) \text{ if } g \equiv 1 \text{ on } O_{|t|}; \quad (\text{P})$$

$$\omega_{\tau_a g} = \omega_g \circ \tau_{-a}, \quad (\text{T})$$

$\tau_a$  the spatial translation. (P) is the Glimm–Jaffe patching construction [1, §III], [2, §3.5–3.6], [6, §4.1]; (T) follows from  $K(\tau_a g) = U_a K(g) U_a^*$  and simplicity (companion M0, Lem. 3.2 — three lines).

**Proposition 2.4** (domain smoothing; cutoff propagation). (i)  $\Omega_g \in D(V(h))$  for every  $g \in \mathcal{G}$  and real  $h \in C_c^\infty$ . (ii)  $\beta_t^g(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|})$  for every  $g$ , with speed  $\leq 1$  independent of  $g$ .

*Proof.* (i) The model-specific smoothing theorem [6, Ch. 7, Thm. 7.4] states that, for every  $t > 0$ , the range of the cutoff semigroup is contained in the domain of each relevant polynomial multiplication operator; in the present notation,  $\text{Ran } e^{-tK(g)} \subseteq D(V(h))$ . (Replacing  $K(g)$  by  $H(g) = K(g) - E(g)$  multiplies the semigroup by the nonzero scalar  $e^{tE(g)}$  and does not change its range.) Since  $K(g)\Omega_g = E(g)\Omega_g$ ,

$$\Omega_g = e^{tE(g)} e^{-tK(g)} \Omega_g \in \text{Ran } e^{-tK(g)} \subseteq D(V(h)).$$

(ii) The interaction automorphism  $\text{Ad } e^{itV(g)}$  has speed zero: for  $A$  generated by Weyl operators supported in  $B \Subset O$  pick  $\chi \in C_c^\infty(O)$ ,  $\chi = 1$  near  $B$ ; then  $V(g) = V(\chi g) + V((1 - \chi)g)$  strongly commuting,  $e^{itV(\chi g)} \in \mathfrak{A}(O)$ ,  $e^{itV((1 - \chi)g)} \in \mathfrak{A}(B)'$ , so  $\text{Ad } e^{itV(g)}(A) \in \mathfrak{A}(O)$ ; extend by strong density. The free dynamics has speed one (Klein–Gordon). The  $n$ -th Trotter approximant of  $\beta_t^g(A)$  then lies in  $\mathfrak{A}(O_{|t|})$ , which is strongly closed, and the Trotter formula converges strongly. [6, Ch. 4]  $\square$

### 3 The source theorems and their cutoff-uniformity

The Euclidean input is [7]. Let  $d\Phi$  be the Gaussian with covariance  $(-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$  and, for  $h$  in the class  $\mathcal{C} = \{0 \leq h \leq 1, \text{ supp } h \text{ compact}\}$  [7, p. 589],

$$dq_h = Z(h)^{-1} e^{-V(h)} d\Phi, \quad V(h) = \lambda \int :\mathcal{P}(\Phi(z)): h(z) dz.$$

The observable class (1.1.5) consists of sharp-time monomials  $Q(x^0) = \int d\vec{x} w(\vec{x}) \prod_i :\Phi(x_i)^{m_i}:$  with  $L^2$  kernels, equal times allowed, normed by (1.1.8)–(1.1.9) (kernel  $L^2$  norms with local-degree weights  $K_2^{2N(\Delta)} N(\Delta)!$  and a single overall prefactor  $K_1$ ; printed forms verified against the page image of p. 594:  $\|Q\| = K_1 [\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)!] \|w\|_2$  and likewise at fixed times). Replacing  $K_2$  by  $\max\{1, K_2\}$  only enlarges the right sides of all source estimates, so we fix, from now on,  $K_2 \geq 1$ .

**Source Fact 3.1** (CE2 = [7, Thm. 1.1.7], p. 594). For localized monomials  $Q_1, Q_2$  with disjoint localization squares separated by a strip of width  $d$  bounded by two parallel lines (any orientation),

$$\left| \int Q_1 Q_2 dq_h - \int Q_1 dq_h \int Q_2 dq_h \right| \leq C_7 e^{-m_1 d} \|Q_1\| \|Q_2\|,$$

with  $m_1, C_7$  independent of  $h, Q_1, Q_2$  (so stated).

**Source Fact 3.2** (CE1 = [7, Thm. 1.1.8], p. 595).  $|\int Q(x^0) dq_h| \leq C_8 \|Q(x^0)\|$ , uniformly in  $h \in \mathcal{C}$ .

**Remark 3.3** (the uniformity audit). The class-uniformity of Source Facts 3.1–3.2 over all of  $\mathcal{C}$  — not merely along  $h \uparrow 1$  — is (a) *printed*: [7, p. 629] states “the estimates of Theorems 1.1.8, 1.1.11, 3.1 and 4.1 are uniform in the space cutoff  $h$ ”; (b) *structural*: the convergence proof is a kernel-norm argument in which  $h$  enters quantitatively only through vertex factors majorised by  $|\lambda h(x)| \leq \lambda$  at first contact, while every  $e^{-V(h)}$  cancels against the normalisation in the final division (their (4.3)–(4.4)), so no stability estimate enters any constant; and (c) the only  $h$ -sensitive analytic prerequisites —  $e^{-\lambda|a_0|K} \leq Z(h) < \infty$  and  $\sup_{h \in \mathcal{C}, \text{supp } h \subseteq K} \|e^{-V(h)}\|_{L^p(d\Phi)} < \infty$  — are proved, self-contained, in the companion audit (m4/r1-check.tex, Lem. 4.2, Thm. 4.5), the two halves of  $0 \leq h \leq 1$  entering complementarily (positivity in the Wick lower bound  $:\mathcal{P}:_c \geq -b(1+c)^{p'}$ , boundedness in the UV kernel estimates). The formerly illegible prefactor of the norm (Gate R2) has been settled by direct inspection of the page image:  $K_1$  enters to the first power and the local weight is  $K_2^{2N(\Delta)}$ ; see m4/r2-check.tex. All constants below use this exact reading.

**Remark 3.4** (conventions lock). The sharp-time restriction of the Euclidean covariance equals  $C: \int \frac{dk_0}{2\pi} (k_0^2 + \mu(k)^2)^{-1} = (2\mu(k))^{-1}$  exactly. Hence sharp-time  $d\Phi$ -Wick ordering coincides, constant by constant, with the  $C$ -Wick ordering used in  $V(h)$ , and sharp-time functionals of  $\Phi(0, \cdot)$  are multiplication operators on  $\mathcal{H}$ . No Wick-mass conversion arises. (Companion m4/t3.tex, Lem. 2.1; m2-audit.tex, Prop. 2.7.)

Throughout §§4–5,  $g \in \mathcal{G}$  is fixed where limits  $T \rightarrow \infty$  are taken; *no uniformity in  $g$  beyond the printed constants is used, and no gap is assumed*. Set

$$h_{T,g} := \mathbf{1}_{[-T,T]} \otimes g \in \mathcal{C}, \quad \psi_s := e^{-sH(g)} \Omega_0, \quad \hat{\psi}_s := \psi_s / \|\psi_s\|, \quad c_g := \langle \Omega_g, \Omega_0 \rangle > 0$$

( $c_g > 0$  by Source Fact 2.2:  $\Omega_g > 0$  a.e.,  $\Omega_0 \equiv 1$ ).

**Lemma 3.5** (slab FKN). *For  $T > 0$ ,  $t \in [0, 2T]$ ,  $s \in [-T, T]$ , and nonnegative measurable functions  $F_1, F_2$  of the time-zero field:*

$$\int F_1(\Phi(-\frac{t}{2}, \cdot)) F_2(\Phi(\frac{t}{2}, \cdot)) dq_{h_{T,g}} = \frac{\langle F_1 \psi_{T-t/2}, e^{-tH(g)} F_2 \psi_{T-t/2} \rangle}{\|\psi_T\|^2}, \quad (1)$$

$$\int F_1(\Phi(s, \cdot)) dq_{h_{T,g}} = \frac{\langle \psi_{T-s}, F_1 \psi_{T+s} \rangle}{\|\psi_T\|^2}, \quad (2)$$

as identities in  $[0, \infty]$ ; energy shifts cancel in the ratios.

*Proof.* For bounded  $F_i$  this is the Feynman–Kac–Nelson formula for  $Z^{-1} e^{-V(h_{T,g})} d\Phi$  (Markov property of  $d\Phi$  along constant-time lines; [12, Thm. III.6, Cor. V.14, Thm. V.15]; [10, Thms. II.14, II.16]; the identification of sharp-time Wick functionals with  $Q$ -space multiplications is Remark 3.4). For bounded nonnegative  $F_i$  both sides are finite. For bounded signed real  $F_i$ , write  $F_i = F_i^+ - F_i^-$  and expand both sides into the same four finite nonnegative terms; bilinearity proves the identities.

For nonnegative measurable  $F_i$ , apply the bounded formula to  $F_i \wedge M$  and let  $M \uparrow \infty$ ; monotone convergence on both sides gives the displayed identities in  $[0, \infty]$ . For the signed unbounded functions used below, define the clipped truncation

$$F^{[M]} := (-M) \vee (F \wedge M).$$

Whenever  $F\psi_a \in \mathcal{H}$ , Hilbert-space convergence follows from

$$\|(F - F^{[M]})\psi_a\|^2 = \int |F - F^{[M]}|^2 |\psi_a|^2 d\phi_0 \longrightarrow 0, \quad |F - F^{[M]}|^2 |\psi_a|^2 \leq |F|^2 |\psi_a|^2.$$

For two path-space factors the pointwise estimate

$$|F_1(\Phi(s_1)) F_2(\Phi(s_2))| \leq \frac{1}{2} (F_1(\Phi(s_1))^2 + F_2(\Phi(s_2))^2)$$

provides a dominated-convergence majorant whenever the two square moments are finite. Source Fact 3.2 supplies precisely those square moments for every later application. Thus the signed identities used below are limits of the bounded signed identities, with no formal manipulation of products of unbounded operators.  $\square$

**Lemma 3.6** (slab vacuum; simplicity only).  $\hat{\psi}_s \rightarrow \Omega_g$  in norm,  $\|\psi_s\| \downarrow c_g$ , and  $\|\psi_{s'}\|/\|\psi_s\| \in [1, 1/c_g]$  for  $0 \leq s' \leq s$ ; the ratio tends to 1 whenever  $s, s' \rightarrow \infty$ .

*Proof.* Set  $P_g = |\Omega_g\rangle\langle\Omega_g|$ . Since  $P_g\Omega_0 = c_g\Omega_g$ ,

$$\psi_s = c_g\Omega_g + e^{-sH(g)}(1 - P_g)\Omega_0,$$

and the two summands are orthogonal. If  $\mu_g^\perp$  is the spectral measure of  $(1 - P_g)\Omega_0$ , then

$$\|e^{-sH(g)}(1 - P_g)\Omega_0\|^2 = \int_{(0,\infty)} e^{-2sE} d\mu_g^\perp(E) \longrightarrow 0$$

by dominated convergence. The measure has no atom at 0 by simplicity alone (Source Fact 2.2); no isolation and hence no gap is used. It follows that  $\psi_s \rightarrow c_g\Omega_g$ ,  $\|\psi_s\| \rightarrow c_g$ , and  $\hat{\psi}_s \rightarrow \Omega_g$ . The spectral formula also shows that  $s \mapsto \|\psi_s\|$  is decreasing. Since  $c_g \leq \|\psi_s\| \leq \|\psi_0\| = 1$ , for  $0 \leq s' \leq s$ ,

$$1 \leq \frac{\|\psi_{s'}\|}{\|\psi_s\|} \leq \frac{1}{c_g}.$$

If both indices tend to infinity, numerator and denominator tend to  $c_g > 0$ , so their ratio tends to 1.  $\square$

## 4 The uniform local interaction moment

**Theorem 4.1** ((M<sub>2</sub>)). *There is  $M = M(\lambda, \mathcal{P}, K_1, K_2, C_8) < \infty$  such that for every  $g \in \mathcal{G}$  and every real  $h \in C_c^\infty(\mathbb{R})$  with  $\|h\|_\infty \leq 1$  and  $\text{supp } h$  contained in an interval of length 2,*

$$\|V(h)\Omega_g\| \leq M.$$

*Explicitly  $M^2 = 9 C_8 \mathfrak{K}(p') \lambda^2 (\sum_n |a_n|)^2$  with  $\mathfrak{K}(p') = K_1 K_2^{8p'} (4p')!$  (per-point degree  $\leq \deg \mathcal{P} = 2p'$ , so the local degrees sum to  $n + n' \leq 4p'$ ).*

*Proof. Step 1 (class membership and norm).* On path space set

$$Q_V := \lambda^2 \sum_{n, n'=0}^{2p'} a_n a_{n'} \iint h(x) h(y) : \Phi(0, x)^n : : \Phi(0, y)^{n'} : dx dy,$$

a sum of equal-time monomials of the class (1.1.5) with  $r = 2$  and kernels  $a_n a_{n'} h \otimes h \in L^2$  — *no Wick recombination occurs*, the product of two individually ordered powers being literally of the form (1.1.5); per-point degrees are  $\leq 2p'$ , admissible by [7, fn. 9]. Let  $J := \{j \in \mathbb{Z} : \|h \mathbf{1}_{[j, j+1)}\|_2 > 0\}$  and decompose  $h = \sum_{j \in J} h \mathbf{1}_{[j, j+1)}$ . An interval of length 2 meets at most three half-open unit cells in positive measure, so  $|J| \leq 3$ , while  $\|h \mathbf{1}_{[j, j+1)}\|_2 \leq \|h\|_\infty \leq 1$ . A localized piece with both points in one square carries  $K_2^{2(n+n')} (n+n')! \leq K_2^{8p'} (4p')!$ ; with points in distinct squares,  $K_2^{2n} n! K_2^{2n'} n'! \leq K_2^{8p'} (4p')!$  — the  $K_2$  exponent  $2 \sum_\Delta N(\Delta) = 2(n+n') \leq 8p'$  is additive and case-independent, and  $(n+n')! \leq (4p')!$  since each degree is at most  $\deg \mathcal{P} = 2p'$ . Summing the  $\leq 9$  pairs and the degrees,  $\|Q_V\| \leq 9 \mathfrak{K}(p') \lambda^2 (\sum |a_n|)^2$ .

*Step 2 (uniform slab bound).* Apply (1) at  $t = 0$  first to  $F_1 = F_2 = |V(h)| \wedge R$ . Both functions are nonnegative. Monotone convergence as  $R \uparrow \infty$ , followed by Source Fact 3.2, gives (Remark 3.4 identifies the sharp-time functional with the operator)

$$\|V(h)\hat{\psi}_T\|^2 = \int Q_V dq_{h_T, g} \leq C_8 \|Q_V\| \leq M^2 \quad \text{for all } T > 0, g \in \mathcal{G},$$

since  $h_{T, g} \in \mathcal{C}$ .

*Step 3 (closedness limit).*  $V(h)$  is multiplication by a real measurable function, hence self-adjoint on its maximal domain and closed. Its graph is a norm-closed linear subspace of  $\mathcal{H} \oplus \mathcal{H}$ , hence is weakly closed. Choose  $T_k \rightarrow \infty$  so that  $\|V(h)\hat{\psi}_{T_k}\|$  tends to its liminf. Step 2 gives a weakly convergent subsequence, say  $V(h)\hat{\psi}_{T_{k_\ell}} \rightharpoonup y$ , while Lemma 3.6 gives  $\hat{\psi}_{T_{k_\ell}} \rightarrow \Omega_g$  strongly. Weak closedness of the graph yields  $\Omega_g \in D(V(h))$  and  $y = V(h)\Omega_g$ . Therefore weak lower semicontinuity gives

$$\|V(h)\Omega_g\| \leq \liminf_{\ell \rightarrow \infty} \|V(h)\hat{\psi}_{T_{k_\ell}}\| = \liminf_{T \rightarrow \infty} \|V(h)\hat{\psi}_T\| \leq M.$$

□

## 5 The uniform gap

Let  $\mathcal{Q}$  be the set of finite products  $Q = \prod_{i=1}^r \varphi_0(f_i)$ ,  $f_i \in C_c^\infty$  real, viewed simultaneously as multiplication operators on  $\mathcal{H}$  and, at Euclidean time  $s$ , as class-(1.1.5) monomials  $Q(s)$  with kernel  $f_1 \otimes \cdots \otimes f_r$  and all  $m_i = 1$ . Write  $\|Q\|_{\text{GJS}}$  for the (1.1.9)-norm of  $Q(s)$ . Let  $I$  be a bounded interval of length  $L$  containing every  $\text{supp } f_i$ , and define

$$J_I := \{j \in \mathbb{Z} : |I \cap [j, j+1)| > 0\}, \quad N_I := \#J_I.$$

If  $j_{\min}$  and  $j_{\max}$  are the extreme elements of  $J_I$ , then points of positive-measure intersection can be chosen arbitrarily close to the two extreme cells' interiors; consequently  $j_{\max} - j_{\min} - 1 < L$ . Since  $N_I = j_{\max} - j_{\min} + 1$  is an integer,

$$N_I \leq \lfloor L \rfloor + 2 \leq L + 2.$$

Decompose  $f_i = \sum_{j \in J_I} f_i \mathbf{1}_{[j, j+1)}$ . For every cell assignment, the product of the local factorials is at most  $r!$ , because  $\prod_{\Delta} N(\Delta)! \leq (\sum_{\Delta} N(\Delta))! = r!$ , and the  $K_2$  exponent is  $2r$ . Moreover Cauchy–Schwarz gives

$$\sum_{j \in J_I} \|f_i \mathbf{1}_{[j, j+1)}\|_2 \leq N_I^{1/2} \left( \sum_{j \in J_I} \|f_i \mathbf{1}_{[j, j+1)}\|_2^2 \right)^{1/2} = N_I^{1/2} \|f_i\|_2.$$

Summing all cell assignments, and applying the same calculation to the  $2r$  factors of  $Q^2$ , yields the exact bounds

$$\|Q\|_{\text{GJS}} \leq K_1 K_2^{2r} r! N_I^{r/2} \prod_{i=1}^r \|f_i\|_2, \quad \|Q^2\|_{\text{GJS}} \leq K_1 K_2^{4r} (2r)! N_I^r \prod_i \|f_i\|_2^2 =: \mathbf{n}(Q)^2. \quad (3)$$

This norm is independent of the common sharp time  $s$ : at every  $s$  all factors occupy one temporal row, and changing  $s$  changes only that common row label. The spatial cell assignments, local degrees, kernel norms, and local weights in the preceding calculation are unchanged; no invariance of a fixed unit lattice under arbitrary continuous translation is asserted.

For  $t > 2$ , the pieces of  $Q(-t/2)$  and of  $\tilde{Q}(t/2)$  are separated by a horizontal strip of width  $\geq t - 2$  (unit lattice rows), so Source Fact 3.1 and bilinearity give the *two-time cluster bound*

$$\left| \int Q(-\tfrac{t}{2}) \tilde{Q}(\tfrac{t}{2}) dq_h - \int Q(-\tfrac{t}{2}) dq_h \int \tilde{Q}(\tfrac{t}{2}) dq_h \right| \leq C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\text{GJS}} \|\tilde{Q}\|_{\text{GJS}}, \quad h \in \mathcal{C}. \quad (4)$$

**Lemma 5.1.** *Fix  $g \in \mathcal{G}$  and  $Q \in \mathcal{Q}$ . Then:*

- (i)  $\sup_T \|Q\hat{\psi}_T\|^2 \leq C_8 \mathbf{n}(Q)^2$ ;  $\Omega_g \in D(Q)$  and  $\|Q\Omega_g\|^2 \leq C_8 \mathbf{n}(Q)^2$ ; for every  $a > 0$ , the clipped truncations  $Q^{[M]} := (-M) \vee (Q \wedge M)$  satisfy  $\|(Q - Q^{[M]})\psi_a\| \rightarrow 0$ ;
- (ii) for every fixed  $s \in \mathbb{R}$ ,  $\int Q(s) dq_{h_{T,g}} \rightarrow \omega_g(Q)$  as  $T \rightarrow \infty$  through  $T > |s|$ ;
- (iii) for every  $t \geq 0$ ,  $\langle Q\Omega_g, e^{-tH(g)} Q\Omega_g \rangle \leq \liminf_{T \rightarrow \infty} \int Q(-\tfrac{t}{2}) Q(\tfrac{t}{2}) dq_{h_{T,g}}$ .

*Proof.* (i). Apply (1) at  $t = 0$  to  $F_1 = F_2 = |Q| \wedge M$ . Monotone convergence and Source Fact 3.2, applied to the class-(1.1.5) observable  $Q^2$ , give

$$\|Q\hat{\psi}_T\|^2 = \int Q(0)^2 dq_{h_{T,g}} \leq C_8 \|Q^2\|_{\text{GJS}} = C_8 \mathbf{n}(Q)^2. \quad (5)$$

Choose a sequence attaining the liminf as  $T \rightarrow \infty$  and repeat the weak-graph argument of Theorem 4.1, Step 3; this proves  $\Omega_g \in D(Q)$  and, by weak lower semicontinuity,  $\|Q\Omega_g\|^2 \leq C_8 \mathbf{n}(Q)^2$ . For fixed  $a > 0$ , (5) with  $T = a$  gives  $Q\psi_a \in \mathcal{H}$ , and dominated convergence on  $Q$ -space gives

$$\|(Q - Q^{[M]})\psi_a\|^2 = \int |Q - Q^{[M]}|^2 |\psi_a|^2 d\phi_0 \rightarrow 0, \quad |Q - Q^{[M]}|^2 |\psi_a|^2 \leq Q^2 |\psi_a|^2.$$

(ii). Apply the bounded signed identity (2) to  $Q^{[M]}$  and pass  $M \rightarrow \infty$  using part (i) on the Hilbert-space side. On path space, the positive and negative parts are integrable because  $\int |Q(s)| dq \leq (\int Q(s)^2 dq)^{1/2}$  and Source Fact 3.2 bounds the square. Thus, for  $T > |s|$ ,

$$\int Q(s) dq_{h_{T,g}} = \left\langle \hat{\psi}_{T-s}, \frac{Q\psi_{T+s}}{\|\psi_T\|} \right\rangle \frac{\|\psi_{T-s}\|}{\|\psi_T\|}. \quad (6)$$

The first vector tends strongly to  $\Omega_g$ , and the final scalar tends to 1 by Lemma 3.6. For the middle vector, write

$$\frac{Q\psi_{T+s}}{\|\psi_T\|} = \frac{\|\psi_{T+s}\|}{\|\psi_T\|} Q\hat{\psi}_{T+s}. \quad (7)$$

The scalar ratio in (7) is at most 1 when  $s \geq 0$ . When  $s < 0$ ,  $T + s < T$  and Lemma 3.6 gives instead

$$1 \leq \frac{\|\psi_{T+s}\|}{\|\psi_T\|} \leq c_g^{-1}.$$

Hence (5) makes the middle vectors bounded in both sign cases. The scalar ratio tends to 1 for fixed  $s$ . Every weakly convergent subsequence of  $Q\hat{\psi}_{T+s}$  has limit  $Q\Omega_g$ : indeed,  $\hat{\psi}_{T+s} \rightarrow \Omega_g$  strongly and the graph of  $Q$  is weakly closed. Thus  $Q\hat{\psi}_{T+s} \rightharpoonup Q\Omega_g$ , and (7) has the same weak limit. Taking the strong-weak pairing in (6) proves  $\int Q(s) dq_{h_{T,g}} \rightarrow \langle \Omega_g, Q\Omega_g \rangle$ .

(iii). Apply (1) first to the bounded signed functions  $Q^{[M]}$  at the two time slices. The path-space products converge in  $L^1$  because

$$|Q^{[M]}(-t/2)Q^{[M]}(t/2)| \leq \frac{1}{2}(Q(-t/2)^2 + Q(t/2)^2),$$

whose integral is finite by Source Fact 3.2. Part (i) gives convergence of the Hilbert-space vectors. Consequently, for  $T > t/2$ ,

$$\int Q(-t/2)Q(t/2) dq_{h_{T,g}} = \|e^{-tH(g)/2}u_T\|^2, \quad u_T := \frac{Q\psi_{T-t/2}}{\|\psi_T\|}. \quad (8)$$

Here

$$\|u_T\| = \frac{\|\psi_{T-t/2}\|}{\|\psi_T\|} \|Q\hat{\psi}_{T-t/2}\| \leq c_g^{-1} C_8^{1/2} \mathbf{n}(Q)$$

by Lemma 3.6 and (5). The ratio tends to 1, and the same weak-graph argument as in (ii) gives  $u_T \rightharpoonup Q\Omega_g$ . Since  $e^{-tH(g)/2}$  is bounded,  $e^{-tH(g)/2}u_T \rightharpoonup e^{-tH(g)/2}Q\Omega_g$ . Weak lower semicontinuity of the norm in (8) now yields

$$\langle Q\Omega_g, e^{-tH(g)}Q\Omega_g \rangle = \|e^{-tH(g)/2}Q\Omega_g\|^2 \leq \liminf_{T \rightarrow \infty} \int Q(-t/2)Q(t/2) dq_{h_{T,g}}.$$

□

**Proposition 5.2** (connected decay). *For real  $Q \in \mathcal{Q}$  and  $\psi_Q := Q\Omega_g - \omega_g(Q)\Omega_g$ :*

$$\langle \psi_Q, e^{-tH(g)}\psi_Q \rangle \leq \hat{C}_Q e^{-m_1 t} \quad (t \geq 0), \quad \hat{C}_Q = \max(C_7 e^{2m_1} \|Q\|_{\text{GJS}}^2, e^{3m_1} \|\psi_Q\|^2).$$

*Proof.*  $\langle \psi_Q, e^{-tH}\psi_Q \rangle = \langle Q\Omega_g, e^{-tH}Q\Omega_g \rangle - \omega_g(Q)^2$  (using  $e^{-tH}\Omega_g = \Omega_g$  and  $Q$  symmetric). For  $t > 2$ , Lemma 5.1(iii), then (4) with  $h = h_{T,g}$  uniformly in  $T$ , then Lemma 5.1(ii) for the disconnected part, give

$$\langle Q\Omega_g, e^{-tH}Q\Omega_g \rangle \leq C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\text{GJS}}^2 + \omega_g(Q)^2,$$

via  $\liminf(a_T + b_T) \leq \sup_T a_T + \lim_T b_T$ . For  $t \leq 3$  use monotonicity of  $t \mapsto \|e^{-tH/2}\psi_Q\|^2$  and  $e^{-m_1 t} \geq e^{-3m_1}$ . □

**Theorem 5.3** (density).  $\text{span}_{\mathbb{C}}\{Q\Omega_g : Q \in \mathcal{Q}\}$  is dense in  $\mathcal{H}$ , for every  $g \in \mathcal{G}$ .

*Proof.* Equivalently (unitarity of  $\psi \mapsto \psi/\Omega_g$  onto  $L^2(d\mu_g)$ ,  $d\mu_g = \Omega_g^2 d\phi_0$ ): polynomials in  $\{\varphi_0(f)\}$  are dense in  $L^2(d\mu_g)$ .

*Exponential moments.* Let  $f$  be real and supported in an interval  $I$ ; use the integer  $N_I$  defined above and set

$$X_f := \varphi_0(f), \quad K_f := K_2^2 N_I^{1/2} \|f\|_2.$$

If  $f = 0$ , then  $X_f = 0$  and all exponential moments are finite; hence assume  $f \neq 0$ , so  $K_f > 0$ . For every  $n \geq 1$ , Lemma 5.1(i), applied to  $Q = X_f^n$ , and (3) give

$$\mathbb{E}_{\mu_g}[X_f^{2n}] = \|X_f^n \Omega_g\|^2 \leq C_8 K_1 K_2^{4n} (2n)! N_I^n \|f\|_2^{2n} = C_8 K_1 K_f^{2n} (2n)!.$$

Since  $e^{s|x|} \leq e^{sx} + e^{-sx} = 2 \cosh(sx)$ , Tonelli's theorem and  $\mathbb{E}_{\mu_g} 1 = 1$  yield, for  $0 \leq s < K_f^{-1}$ ,

$$\mathbb{E}_{\mu_g} e^{s|X_f|} \leq 2 \left( 1 + C_8 K_1 \sum_{n=1}^{\infty} (s K_f)^{2n} \right) \leq 2 \max\{1, C_8 K_1\} \sum_{n=0}^{\infty} (s K_f)^{2n} < \infty.$$

Thus  $K_1$  affects the prefactor but not the nonzero radius of exponential integrability.

*Orthogonality kills conditional expectations.* Suppose  $\zeta \in L^2(d\mu_g)$  is orthogonal to every cylinder polynomial. Fix real  $e_1, \dots, e_k \in C_c^\infty$ , put  $X_j = \varphi_0(e_j)$ , and choose  $r_j > 0$  with  $\mathbb{E}_{\mu_g} e^{r_j |X_j|} < \infty$  by the preceding paragraph. Choose  $\sigma > 0$  so that  $2k\sigma < \min_j r_j$ . For  $|\text{Re } z_j| < \sigma$  define

$$F(z_1, \dots, z_k) := \mathbb{E}_{\mu_g} \left[ \bar{\zeta} \exp \left( \sum_{j=1}^k z_j X_j \right) \right].$$

Cauchy–Schwarz followed by Hölder with  $k$  equal exponents gives the explicit dominator

$$\begin{aligned} \mathbb{E}_{\mu_g} \left[ |\zeta| e^{\sum_j \text{Re } z_j X_j} \right] &\leq \|\zeta\|_2 \left( \mathbb{E}_{\mu_g} e^{2\sigma \sum_j |X_j|} \right)^{1/2} \\ &\leq \|\zeta\|_2 \prod_{j=1}^k \left( \mathbb{E}_{\mu_g} e^{2k\sigma |X_j|} \right)^{1/(2k)} < \infty. \end{aligned}$$

On every closed smaller polystrip, derivatives are dominated by the same expression with a slightly larger exponent, because for  $a > 0$  and  $n \in \mathbb{N}$ ,  $|x|^n \leq n! a^{-n} e^{a|x|}$ . Hence differentiation under the integral is justified and  $F$  is jointly analytic. Every derivative at the origin equals the inner product of  $\zeta$  with a cylinder monomial and therefore vanishes. Applying the one-variable identity theorem successively in  $z_1, \dots, z_k$  gives  $F \equiv 0$  on the connected polystrip, in particular on  $(i\mathbb{R})^k$ .

Define the finite complex Borel measure on  $\mathbb{R}^k$

$$\nu(B) := \mathbb{E}_{\mu_g} \left[ \bar{\zeta} \mathbf{1}_{\{(X_1, \dots, X_k) \in B\}} \right].$$

Its Fourier transform is  $F(it_1, \dots, it_k) = 0$ . Uniqueness of Fourier transforms of finite measures gives  $\nu = 0$ , which is equivalent to  $\mathbb{E}_{\mu_g}[\zeta \mid \Sigma_k] = 0$  for  $\Sigma_k := \sigma(X_1, \dots, X_k)$ .

*Martingale exhaustion.* Choose a sequence  $(e_j) \subset C_c^\infty(\mathbb{R}; \mathbb{R})$  dense in real  $L^2(\mathbb{R})$  and let  $\Sigma_k = \sigma(\varphi_0(e_1), \dots, \varphi_0(e_k))$ . The bounded free covariance  $C = (2\sqrt{-\partial_x^2 + m_0^2})^{-1}$  gives

$$\mathbb{E}_{\phi_0} |\varphi_0(f) - \varphi_0(e)|^2 = \langle f - e, C(f - e) \rangle \leq \|C\| \|f - e\|_2^2.$$

For any  $f \in L^2(\mathbb{R}; \mathbb{R})$ , an approximating sequence has a subsequence converging  $\phi_0$ -almost surely and therefore  $\mu_g$ -almost surely, because  $d\mu_g = \Omega_g^2 d\phi_0 \ll d\phi_0$ . Thus every cylinder

coordinate is measurable modulo  $\mu_g$ -null sets with respect to  $\Sigma_\infty := \bigvee_k \Sigma_k$ , so  $\Sigma_\infty$  is the full cylinder sigma-algebra. The  $L^2$  martingale convergence theorem now gives

$$\zeta = \mathbb{E}_{\mu_g}[\zeta \mid \Sigma_\infty] = \lim_{k \rightarrow \infty} \mathbb{E}_{\mu_g}[\zeta \mid \Sigma_k] = 0.$$

□

**Theorem 5.4** ((G)). *For every  $g \in \mathcal{G}$ :  $\text{spec } H(g) \subseteq \{0\} \cup [m_1, \infty)$ , and 0 is simple.*

*Proof.* Let  $D = \{\psi_Q : Q \in \mathcal{Q} \text{ real}\}$ . The complex span of  $D \cup \{\Omega_g\}$  equals the complex span of  $\{Q\Omega_g : Q \in \mathcal{Q}\}$  and is dense by Theorem 5.3. Fix  $0 \leq \beta < m_1$  and write  $P_\beta := \mathbf{1}_{[0, \beta]}(H(g))$ . For  $\psi_Q \in D$ , the spectral theorem and Proposition 5.2 give, for every  $t \geq 0$ ,

$$e^{-\beta t} \|P_\beta \psi_Q\|^2 \leq \langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \leq \widehat{C}_Q e^{-m_1 t}.$$

Multiplication by  $e^{\beta t}$  and passage to  $t \rightarrow \infty$  yield  $P_\beta \psi_Q = 0$ , because  $m_1 - \beta > 0$ . On the other hand,  $P_\beta \Omega_g = \Omega_g$  because  $0 \in [0, \beta]$ . Thus  $P_\beta$  maps a dense subspace into  $\mathbb{C}\Omega_g$ . Continuity of the bounded projection extends this containment to all of  $\mathcal{H}$ , and  $P_\beta \Omega_g = \Omega_g$  shows

$$P_\beta = |\Omega_g\rangle\langle\Omega_g| \quad (0 \leq \beta < m_1).$$

If  $\text{spec } H(g)$  met  $(0, m_1)$ , an open interval  $J \Subset (0, m_1)$  meeting the spectrum would have a nonzero spectral projection. Choosing  $\beta < m_1$  above  $\sup J$  would give

$$0 \neq \mathbf{1}_J(H(g)) \leq P_\beta - |\Omega_g\rangle\langle\Omega_g| = 0,$$

a contradiction. Hence  $\text{spec } H(g) \subseteq \{0\} \cup [m_1, \infty)$ ; simplicity of the zero eigenvalue is Source Fact 2.2. □

**Remark 5.5.** The constants  $\widehat{C}_Q$  of Proposition 5.2 are in fact  $g$ -uniform, but this is never used: the spectral argument needs only the *rate* to be  $g$ -independent. Accordingly (M<sub>2</sub>) and (G) are logically independent consequences of the same source pair. No gap, and no isolation of  $E(g)$ , was assumed at any point of §§3–5; isolation with the uniform bound  $m_1$  is part of the conclusion of Theorem 5.4.

## 6 The Cauchy step (LPPL)

Throughout this section, (G) and (M) denote the statements supplied by Theorems 5.4 and 4.1:  $\text{spec } H(g) \subseteq \{0\} \cup [\gamma, \infty)$  for every  $g \in \mathcal{G}$  with  $\gamma := m_1$ , and  $\sup_g \|V(h)\Omega_g\| \leq M$  for real  $h \in C_c^\infty$  with  $\|h\|_\infty \leq 1$  and support in an interval of length 2.

### 6.1 Affine-path regularity from FKN

**Proposition 6.1** (path regularity). *Let  $g, g' \in \mathcal{G}$  be equal to one on a common nonempty interval,  $g_s = (1-s)g + sg'$ ,  $v = g' - g$ , and*

$$K_s = K(g_s), \quad E_s = E(g_s), \quad H_s = K_s - E_s, \quad P_s = |\Omega_s\rangle\langle\Omega_s|, \quad Q_s = 1 - P_s.$$

*Under (G),  $s \mapsto P_s$  is norm-continuous on  $[0, 1]$  and  $C^1$  on  $(0, 1)$ ; locally on  $(0, 1)$  normalized ground states can be chosen  $C^1$  in the parallel gauge, and then*

$$\dot{E}_s = \langle \Omega_s, V(v)\Omega_s \rangle, \quad \dot{\Omega}_s = -H_s^{-1} Q_s V(v) \Omega_s. \quad (9)$$

*In particular  $s \mapsto \omega_{g_s}(A) = \text{Tr}(P_s A)$  is continuous on  $[0, 1]$  for every  $A \in B(\mathcal{H})$ .*

*Proof.* Fix  $T > 0$ . On the free Euclidean path space  $(\mathcal{S}'(\mathbb{R}^2), d\mu_0)$ ,  $d\mu_0 = d\Phi$ , let  $J_t$  denote the time- $t$  slice embedding,  $(J_t f)(\Phi) = f(\Phi(t, \cdot))$  for bounded cylinder  $f$ , and set

$$\mathcal{U}_s = \lambda \int_0^T \int_{\mathbb{R}} g_s(x) : \mathcal{P}(\Phi(t, x)) : dx dt, \quad w_s = e^{-\mathcal{U}_s}, \quad \mathcal{U}_v = \mathcal{U}_1 - \mathcal{U}_0.$$

The Feynman–Kac–Nelson formula reads  $\langle f, e^{-TK_s} h \rangle = \int \overline{J_T f} (J_0 h) w_s d\mu_0$  for bounded time-zero  $f, h$  [12, Cor. V.14, Thm. V.15]; [10, Thm. II.16].

*Time-slice product estimate.* Put  $q_T = 2/(1 - e^{-m_0 T})$  and  $r_T = q_T/(q_T - 1) = 2/(1 + e^{-m_0 T})$ , so  $p_T := 2/r_T = 1 + e^{-m_0 T}$  has conjugate  $p'_T = 1 + e^{m_0 T}$  and  $(p'_T - 1)/(p_T - 1) = e^{2m_0 T}$  — exactly the Nelson hypercontractivity condition for the free Markov semigroup  $P_T = e^{-TH_0}$  on the time-zero Gaussian  $\nu_0$ :  $\|P_T F\|_{p'_T} \leq \|F\|_{p_T}$  [12, Thms. I.17, III.17]. The Markov property and Hölder then give

$$\|(J_T f)(J_0 h)\|_{r_T}^{r_T} = \int |h|^{r_T} P_T(|f|^{r_T}) d\nu_0 \leq \| |h|^{r_T} \|_{p_T} \| |f|^{r_T} \|_{p_T} = \|h\|_2^{r_T} \|f\|_2^{r_T},$$

using  $r_T p_T = 2$ ; hence

$$\|(J_T f)(J_0 h)\|_{r_T} \leq \|f\|_2 \|h\|_2. \quad (10)$$

*Weight regularity.*  $\mathcal{U}_v \in L^p(\mu_0)$  for every  $p < \infty$ , and since  $g, g' \geq 0$  and  $\mathcal{P}$  is semi-bounded,  $e^{-a\mathcal{U}_0}, e^{-a\mathcal{U}_1} \in L^1(\mu_0)$  for every  $a < \infty$  [12, Thm. V.7] (this is the  $h$ -uniform stability of Remark 3.3(c)). By  $\mathcal{U}_s = (1 - s)\mathcal{U}_0 + s\mathcal{U}_1$  and Hölder,

$$\|w_s\|_a^a \leq \left( \int e^{-a\mathcal{U}_0} d\mu_0 \right)^{1-s} \left( \int e^{-a\mathcal{U}_1} d\mu_0 \right)^s, \quad (11)$$

so  $\sup_{0 \leq s \leq 1} \|w_s\|_a < \infty$ . If  $s_n \rightarrow s$  then  $\mathcal{U}_{s_n} \rightarrow \mathcal{U}_s$  in every  $L^p$ , so  $w_{s_n} \rightarrow w_s$  in probability; (11) at a larger exponent makes  $(|w_{s_n}|^a)$  uniformly integrable, and Vitali gives  $\|w_{s_n} - w_s\|_a \rightarrow 0$ . Pointwise on path space,

$$\frac{w_{s+h} - w_s}{h} = -\mathcal{U}_v \int_0^1 w_{s+\theta h} d\theta,$$

so for  $0 < s < 1$ , choosing  $p, a < \infty$  with  $q_T^{-1} = p^{-1} + a^{-1}$ ,

$$\left\| \frac{w_{s+h} - w_s}{h} + \mathcal{U}_v w_s \right\|_{q_T} \leq \|\mathcal{U}_v\|_p \sup_{0 \leq \theta \leq 1} \|w_{s+\theta h} - w_s\|_a \rightarrow 0 :$$

$s \mapsto w_s$  is continuous  $[0, 1] \rightarrow L^{q_T}$  and  $C^1(0, 1) \rightarrow L^{q_T}$  with continuous derivative  $-\mathcal{U}_v w_s$ .

*Transfer to the semigroup.* Since  $1/r_T + 1/q_T = 1$ , FKN and (10) give  $\|e^{-TK_s} - e^{-TK_r}\| \leq \|w_s - w_r\|_{q_T}$  (first on bounded  $f, h$ , then by  $L^2$  density), and the same estimate for difference quotients:  $s \mapsto L_s := e^{-TK_s}$  is norm-continuous on  $[0, 1]$ , norm- $C^1$  on  $(0, 1)$ .

*Spectral consequences.* The largest spectral value of  $L_s$  is the simple eigenvalue  $\lambda_s = e^{-TE_s} = \|L_s\|$ , hence continuous; by (G),  $\text{spec } L_s \setminus \{\lambda_s\} \subseteq [0, e^{-T\gamma} \lambda_s]$ , a relative separation locally bounded below. The Riesz integral over a circle about  $\lambda_s$  yields norm continuity of  $P_s$  on  $[0, 1]$  and  $C^1$  on  $(0, 1)$ . Near  $s_0 \in (0, 1)$ ,  $\tilde{\Omega}_s = P_s \Omega_{s_0} / \|P_s \Omega_{s_0}\|$  is  $C^1$ ; a  $C^1$  phase puts it in the parallel gauge  $\langle \Omega_s, \dot{\Omega}_s \rangle = 0$ ; and  $E_s = -T^{-1} \log \langle \Omega_s, L_s \Omega_s \rangle$  is  $C^1$ .

*Identification of the derivatives.* On the common core  $\mathcal{C} = C^\infty(H_0)$ ,  $K_s \xi = K_{s_0} \xi + (s - s_0)V(v)\xi$  [12, Thm. V.12(a)]. Differentiating  $\langle K_s \xi, \Omega_s \rangle = E_s \langle \xi, \Omega_s \rangle$  at fixed  $\xi \in \mathcal{C}$ , using  $\Omega_s \in D(V(v))$  (Proposition 2.4) and self-adjointness of  $V(v)$ ,

$$\langle H_s \xi, \dot{\Omega}_s \rangle = \langle \xi, (\dot{E}_s - V(v)) \Omega_s \rangle \quad (\xi \in \mathcal{C}).$$

$\mathcal{C}$  is a core for  $H_s$ , so  $\dot{\Omega}_s \in D(H_s)$  and  $H_s \dot{\Omega}_s = (\dot{E}_s - V(v)) \Omega_s$ ; pairing with  $\Omega_s$  gives the first identity of (9), and the parallel gauge with  $\|H_s^{-1} Q_s\| \leq \gamma^{-1}$  gives the second. Finally  $|\text{Tr}((P_s - P_r)A)| \leq \|P_s - P_r\|_1 \|A\| \leq 2\|P_s - P_r\| \|A\|$  (rank  $\leq 2$ ), which yields the endpoint continuity assertion.  $\square$

## 6.2 The exact spectral filter

**Lemma 6.2** (filter). *For every  $\gamma > 0$  there is  $W_\gamma \in L^1(\mathbb{R})$  with  $\widehat{W}_\gamma(E) := \int W_\gamma(t)e^{itE}dt$  satisfying*

$$\widehat{W}_\gamma(0) = 0, \quad \widehat{W}_\gamma(E) = \frac{1}{E} \quad (|E| \geq \gamma), \quad \int (1 + |t|)^q |W_\gamma(t)| dt < \infty \quad (\forall q \geq 0).$$

*Consequently, if  $H \geq 0$ ,  $\text{spec } H \subseteq \{0\} \cup [\gamma, \infty)$ ,  $Q = 1 - P_{\ker H}$ , then  $H^{-1}Q = \int W_\gamma(t)e^{itH}Q dt$  as a norm-convergent Bochner integral. Write  $T_\gamma(r) := \int_{|t|>r} |W_\gamma| dt \leq C_{q,\gamma}(1+r)^{-q}$ .*

*Proof.* Pick an even  $\chi \in C^\infty(\mathbb{R}; [0, 1])$ ,  $\chi = 0$  on  $[-\gamma/2, \gamma/2]$ ,  $\chi = 1$  off  $(-\gamma, \gamma)$ , and set  $\widehat{W}_\gamma(E) = \chi(E)/E$  ( $= 0$  at  $E = 0$ ): a smooth odd function whose  $k$ -th derivative is integrable for every  $k \geq 1$  (smooth on the compact transition region,  $O(E^{-k-1})$  outside). Define  $W_\gamma(t) = \frac{1}{2\pi} \int e^{-itE} \widehat{W}_\gamma(E) dE$  as an oscillatory integral; by oddness,  $W_\gamma(t) = -\frac{i}{\pi} \int_0^\infty \sin(tE) \chi(E) E^{-1} dE$ , which is uniformly bounded for  $0 < |t| \leq 1$  (absolute bound on  $[0, \gamma]$ ; substitution  $u = |t|E$  and Dirichlet's test on  $[\gamma, \infty)$ ). For  $|t| \geq 1$ ,  $k$  integrations by parts give  $|W_\gamma(t)| \leq \|\widehat{W}_\gamma^{(k)}\|_{L^1}/(2\pi|t|^k)$ , all boundary terms vanishing;  $k > q + 1$  gives the moment bound and  $W_\gamma \in L^1$ . Fourier inversion in  $\mathcal{S}'$  plus continuity of the transform of an  $L^1$  function give  $\mathcal{F}W_\gamma = \widehat{W}_\gamma$  pointwise, and the spectral theorem yields  $\int W_\gamma(t)e^{itH}Q dt = \widehat{W}_\gamma(H)Q = H^{-1}Q$ , norm-convergent since  $W_\gamma \in L^1$ ,  $\|e^{itH}Q\| \leq 1$ . The tail bound is Markov's inequality.  $\square$

## 6.3 Clustering with one unbounded local observable

**Lemma 6.3** (spectral separation). *Let*

$$F(t) = \int_{[\gamma, \infty)} e^{itE} d\rho(E), \quad G(t) = \int_{(-\infty, -\gamma]} e^{itE} d\sigma(E),$$

*with  $|\rho|(\mathbb{R}), |\sigma|(\mathbb{R}) \leq M_0$ . If  $F = G$  on  $(-R, R)$  and  $\gamma R \geq 1$ , then  $|F(0)| \leq 2M_0 e^{-\gamma R/2}$ .*

*Proof.* For  $a, \varepsilon > 0$  let  $f_{a,\varepsilon}(t) = \frac{1}{2\pi i} e^{-t^2/2a} (t - i\varepsilon)^{-1}$ ; residues and Gaussian convolution give  $\check{f}_{a,\varepsilon}(E) = \int_0^\infty \sqrt{a/2\pi} e^{-a(E-E')^2/2} e^{-\varepsilon E'} dE' \in [0, 1]$ , with  $\check{f}_{a,\varepsilon} \rightarrow \Phi(\sqrt{a}E)$  as  $\varepsilon \downarrow 0$  ( $\Phi$  = normal distribution function), and  $\int_{|t| \geq R} |f_{a,\varepsilon}| \leq \frac{a}{\pi R^2} e^{-R^2/2a}$ . Fubini,  $F = G$  on  $(-R, R)$ , and the variation bounds give  $|\int \Phi(\sqrt{a}E) d\rho - \int \Phi(\sqrt{a}E) d\sigma| \leq \frac{2M_0 a}{\pi R^2} e^{-R^2/2a}$  after  $\varepsilon \downarrow 0$ . With  $1 - \Phi(x), \Phi(-x) \leq \frac{1}{2} e^{-x^2/2}$  and the support conditions,  $|F(0)| \leq M_0 e^{-a\gamma^2/2} + \frac{2M_0 a}{\pi R^2} e^{-R^2/2a}$ ;  $a = R/\gamma$  and  $\gamma R \geq 1$  give the claim.  $\square$

**Lemma 6.4** (one unbounded argument). *Let  $H \geq 0$  have simple normalized ground state  $\Omega$ ,  $\text{spec } H \subseteq \{0\} \cup [\gamma, \infty)$ , and dynamics  $\beta_t = \text{Ad } e^{itH}$  with exact finite propagation for the net  $\mathfrak{A}(\cdot)$ . Let  $B \subset \mathbb{R}$  be compact,  $X = X^*$  affiliated with  $\mathfrak{A}(U)$  for every bounded open  $U \supseteq B$ ,  $\Omega \in D(X)$ , and  $A \in \mathfrak{A}(O)$ . If  $R := \text{dist}(B, O) > 0$  and  $\gamma R \geq 1$ , then*

$$|\langle X\Omega, A\Omega \rangle - \langle X\Omega, \Omega \rangle \langle \Omega, A\Omega \rangle| \leq 2 \|X\Omega\| \|A\| e^{-\gamma R/2}.$$

*Proof.* Set  $P_0 = |\Omega\rangle\langle\Omega|$ ,  $F(t) = \langle X\Omega, (e^{itH} - P_0)A\Omega \rangle$ ,  $G(t) = \langle A^*\Omega, (e^{-itH} - P_0)X\Omega \rangle$ : complex spectral measures supported in  $[\gamma, \infty)$ ,  $(-\infty, -\gamma]$ , variations  $\leq M_0 = \|X\Omega\| \|A\|$ . For  $|t| < R$  choose a bounded open  $U \supseteq B$  disjoint from  $O_{|t|}$ ; finite propagation puts  $\beta_t(A) \in \mathfrak{A}(O_{|t|}) \subseteq \mathfrak{A}(U)'$ , and affiliation makes every element of  $\mathfrak{A}(U)'$  preserve  $D(X)$  and commute with  $X$  there. With  $e^{-itH}\Omega = \Omega$ ,

$$\langle X\Omega, e^{itH} A\Omega \rangle = \langle \Omega, X\beta_t(A)\Omega \rangle = \langle \Omega, \beta_t(A)X\Omega \rangle = \langle A^*\Omega, e^{-itH} X\Omega \rangle,$$

and the  $P_0$  terms agree since  $\langle X\Omega, \Omega \rangle \in \mathbb{R}$ . So  $F = G$  on  $(-R, R)$ ; Lemma 6.3 bounds  $F(0)$ , which is the displayed left side.  $\square$

## 6.4 The state derivative

**Lemma 6.5** (state derivative). *Assume (G). In the setting of Proposition 6.1, for  $0 < s < 1$  and self-adjoint  $A \in B(\mathcal{H})$ , with  $V := V(g' - g)$ :*

$$\frac{d}{ds} \omega_{g_s}(A) = -2 \operatorname{Re} \int_{\mathbb{R}} W_{\gamma}(t) \left[ \langle V \Omega_s, \beta_t^{g_s}(A) \Omega_s \rangle - \langle V \Omega_s, \Omega_s \rangle \omega_{g_s}(A) \right] dt, \quad (12)$$

*absolutely convergent.*

*Proof.* By (9) and the parallel gauge,  $\frac{d}{ds} \omega_{g_s}(A) = 2 \operatorname{Re} \langle \dot{\Omega}_s, A \Omega_s \rangle = -2 \operatorname{Re} \langle V \Omega_s, H_s^{-1} Q_s A \Omega_s \rangle$  (all pairings defined by Proposition 2.4). Insert the filter (Lemma 6.2):  $H_s^{-1} Q_s A \Omega_s = \int W_{\gamma}(t) e^{itH_s} Q_s A \Omega_s dt$ ; since  $e^{itH_s} A \Omega_s = \beta_t^{g_s}(A) \Omega_s$  and  $P_s A \Omega_s = \omega_{g_s}(A) \Omega_s$ , the integrand is the bracket of (12). Absolute convergence:  $|\langle V \Omega_s, e^{itH_s} Q_s A \Omega_s \rangle| \leq \|V \Omega_s\| \|A\|$  and  $W_{\gamma} \in L^1$ .  $\square$

## 6.5 The cell estimate and the Cauchy theorem

Fix  $\rho \in C_c^{\infty}((-1, 1))$ ,  $0 \leq \rho \leq 1$ ,  $\rho > 0$  on  $[-\frac{1}{2}, \frac{1}{2}]$ ; put  $D(x) = \sum_k \rho(x - k) > 0$  (smooth, 1-periodic) and  $\eta_j = \rho(\cdot - j)/D$ : a smooth partition of unity with  $0 \leq \eta_j \leq 1$ ,  $\operatorname{supp} \eta_j \subset (j - 1, j + 1)$ ,  $\sum_j \eta_j = 1$ . For  $v = g' - g$  set  $h_j = \eta_j v$ ,  $V_j = V(h_j)$ ; finitely many are nonzero,  $V(v)\xi = \sum_j V_j \xi$  on common domains, and — since  $|v| \leq 1$ ,  $\|h_j\|_{\infty} \leq 1$ ,  $\operatorname{supp} h_j \subset (j - 1, j + 1)$  — hypothesis (M) gives

$$\|V_j \Omega_s\| \leq M \quad (0 \leq s \leq 1). \quad (13)$$

**Proposition 6.6** (one path). *Assume (G) and (M). Let  $O$  be a bounded interval and  $g, g' \in \mathcal{G}$  with  $g = g' = 1$  on  $O_d$ ,  $d \geq d_0 := 4 + 2/\gamma$ . Then*

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq 32M \sum_{n \geq [d]} \left( \|W_{\gamma}\|_1 e^{-\gamma n/4} + T_{\gamma}(n/2) \right). \quad (14)$$

*Proof.* It suffices to bound  $|\omega_g(A) - \omega_{g'}(A)|$  for self-adjoint  $A \in \mathfrak{A}(O)$ ,  $\|A\| \leq 1$  ( $\omega_g - \omega_{g'}$  is hermitian). Since  $0 \leq g, g' \leq 1$ ,  $g_s(x) = 1 \Leftrightarrow g(x) = g'(x) = 1$ ; hence  $g_s \in \mathcal{G}$  and  $g_s = 1$  on  $O_d$  for all  $s$ , so (G), (M) and Proposition 6.1 apply along the whole path. Every nonzero  $h_j$  is supported at distance  $r_j := \operatorname{dist}(B_j, O) \geq d$  from  $O$ ,  $B_j := \operatorname{supp} h_j$ .

Fix  $s \in (0, 1)$ ,  $j$ , and set  $C_{j,s}(t) := \langle V_j \Omega_s, \beta_t^{g_s}(A) \Omega_s \rangle - \langle V_j \Omega_s, \Omega_s \rangle \omega_{g_s}(A)$ . For  $|t| \leq r_j/2$ : Proposition 2.4(ii) gives  $\beta_t^{g_s}(A) \in \mathfrak{A}(O_{|t|})$  with  $\operatorname{dist}(B_j, O_{|t|}) \geq r_j/2$ , and  $\gamma r_j/2 \geq 1$  (as  $r_j \geq d \geq 2/\gamma$ ); Lemma 6.4 with  $X = V_j$  (affiliation and  $\Omega_s \in D(V_j)$  by §2 and Proposition 2.4(i)) and (13) give

$$|C_{j,s}(t)| \leq 2M e^{-\gamma r_j/4}. \quad (15)$$

For all  $t$ , Cauchy–Schwarz gives  $|C_{j,s}(t)| \leq 2M$ . Substituting  $V = \sum_j V_j$  into (12) and splitting each  $t$ -integral at  $|t| = r_j/2$ ,

$$\left| \frac{d}{ds} \omega_{g_s}(A) \right| \leq 4M \sum_j \left( \|W_{\gamma}\|_1 e^{-\gamma r_j/4} + T_{\gamma}(r_j/2) \right).$$

*Shell count.* Write  $O = (a, b)$ . Since  $d \geq 4$  and  $B_j \subset [j - 1, j + 1]$ , a nonzero  $B_j$  lies on one side of  $O_d$ ; if on the right with  $r_j \in [n, n + 1]$ , compactness gives  $x_j \in B_j$  with  $x_j - b = r_j$ , and  $|j - x_j| \leq 1$  forces  $j \in [b + n - 1, b + n + 2]$  — at most four integers; same on the left: at most eight  $j$  per shell. Both  $r \mapsto e^{-\gamma r/4}$  and  $r \mapsto T_{\gamma}(r/2)$  are decreasing and  $r_j \geq d$ , so the sum over  $j$  is at most  $8 \sum_{n \geq [d]} (\dots)$ , giving the bound  $32M \sum_{n \geq [d]} (\dots)$  for the derivative. Integrate over  $[\varepsilon, 1 - \varepsilon]$  and use the endpoint norm continuity of  $P_s$  (Proposition 6.1), then  $\varepsilon \downarrow 0$  and the supremum over the unit ball.  $\square$

Define

$$\Psi_\gamma(d) := \begin{cases} 32M \sum_{n \geq \lfloor d \rfloor} \left( \|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \right), & d \geq d_0, \\ \max\{2, \Psi_\gamma(d_0)\}, & 0 \leq d < d_0. \end{cases} \quad (16)$$

**Lemma 6.7** (decay of the rate function). *For every  $N \geq 1$  there is  $C_{N,\gamma,M}$  with  $\Psi_\gamma(d) \leq C_{N,\gamma,M}(1+d)^{-N}$ .*

*Proof.* The exponential series has an exponentially decaying tail; for the other,  $T_\gamma(r) \leq r^{-q} \int |t|^q |W_\gamma|$  (Lemma 6.2), take  $q = N + 2$  and sum  $n^{-q}$ .  $\square$

**Theorem 6.8** (Cauchy/LPPL). *Assume (G) and (M) — supplied here by Theorems 5.4 and 4.1 with  $\gamma = m_1$ . Then there is a state  $\omega_\infty$  on  $\mathfrak{A}$  with  $\lim_{g \in \mathcal{G}} \|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| = 0$  for every bounded interval  $O$ ; for  $g \equiv 1$  on  $O_d$ ,*

$$\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \Psi_{m_1}(d);$$

*and  $\omega_\infty|_{\mathfrak{A}(O)}$  is normal for every  $O$ .*

*Proof.* Fix  $O, \varepsilon > 0$ ; by Lemma 6.7 pick  $d \geq d_0$  with  $\Psi_{m_1}(d) < \varepsilon$ , and  $g_0 \in \mathcal{G}$  with  $g_0 \equiv 1$  on  $O_d$ . If  $g, g' \succcurlyeq g_0$  then both equal 1 on  $O_d$  (in particular on a common interval), so Proposition 6.6 applies directly to the pair and gives  $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq \Psi_{m_1}(d) < \varepsilon$ . Hence  $(\omega_g|_{\mathfrak{A}(O)})_{g \in \mathcal{G}}$  is Cauchy in  $\mathfrak{A}(O)^*$ ; the limits are compatible under isotony and extend by norm continuity to a state  $\omega_\infty$  on  $\mathfrak{A}$ . The rate follows by fixing  $g \equiv 1$  on  $O_d$  and letting  $g'$  run through the cofinal tail of cutoffs equal to 1 on  $O_d$ ; for  $d < d_0$  the bound is trivial. Normality: normal functionals form a norm-closed subspace of  $\mathfrak{A}(O)^*$  — if  $0 \leq A_i \uparrow A$  in  $\mathfrak{A}(O)$  and  $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| < \delta$ , then  $0 \leq \omega_\infty(A) - \omega_\infty(A_i) \leq 2\delta\|A\| + \omega_g(A - A_i)$ ; let  $i \uparrow$  (normality of  $\omega_g$ ), then  $\delta \downarrow 0$ .  $\square$

## 7 The limit is a ground state; invariances

We import from companion M0 (proofs there are complete and short):

**Proposition 7.1** (exact spectral passage; M0). (i) *If  $(g_n) \subseteq \mathcal{G}$  satisfies  $d(g_n, O) \rightarrow \infty$  for every bounded  $O$ , and  $\omega_{g_n} \rightarrow \omega$  in norm on every  $\mathfrak{A}(O)$ , then  $\omega$  is a ground state of  $(\mathfrak{A}, \alpha)$  in the spectral sense:  $\omega \circ \alpha_t = \omega$ , the function  $t \mapsto \omega(A\alpha_t(B))$  is continuous and bounded for all  $A, B \in \mathfrak{A}$ , and  $\int f(t)\omega(A\alpha_t(B))dt = 0$  for every  $f \in L^1(\mathbb{R})$  whose transform  $\check{f}(E) = \int f(t)e^{itE}dt$  vanishes on  $[0, \infty)$ .* (ii)  $\omega_{\tau_a g} = \omega_g \circ \tau_{-a}$  for all  $a \in \mathbb{R}, g \in \mathcal{G}$ . (iii) *For even  $\mathcal{P}$ ,  $\omega_g \circ \theta = \omega_g$  for every  $g$ , where  $\theta = \text{Ad } \Gamma(-1)$ .*

*Proof.* (ii)  $U_a K(g) U_a^* = K(\tau_a g)$  (the free part commutes with translations; the interaction transforms covariantly), so by simplicity (Source Fact 2.2)  $\Omega_{\tau_a g} = U_a \Omega_g$  up to a phase, which cancels in the state.

(iii)  $\Theta := \Gamma(-1)$  is unitary, self-adjoint, commutes with  $H_0$ , and for even  $\mathcal{P}$  satisfies  $\Theta V(g) \Theta = V(g)$ ; hence  $\Theta \Omega_g = c \Omega_g$  with  $c \in \{\pm 1\}$  by simplicity and  $\Theta^2 = 1$ . In  $Q$ -space  $(\Theta\psi)(q) = \psi(-q)$ , so  $\Theta \Omega_g > 0$  a.e. together with  $\Omega_g > 0$  forces  $c = +1$ .

(i) *Exact identity.* Fix a bounded interval  $O, A, B \in \mathfrak{A}(O)$ ,  $M_0 := \|A\| \|B\|$ , and  $g \in \mathcal{G}$ ; let  $\rho_g(\cdot) := \langle A^* \Omega_g, P_{H(g)}(\cdot) B \Omega_g \rangle$ , a complex measure with  $\text{supp } \rho_g \subseteq \text{spec } H(g) \subseteq [0, \infty)$  and  $|\rho_g|(\mathbb{R}) \leq M_0$ , and  $G_g(t) := \int e^{itE} d\rho_g(E)$ . For  $|t| < d(g, O)$ , the cutoff equals one on  $O_{|t|}$ , so by (P) and  $e^{-itH(g)} \Omega_g = \Omega_g$ ,

$$\omega_g(A\alpha_t(B)) = \langle \Omega_g, A e^{itH(g)} B e^{-itH(g)} \Omega_g \rangle = \langle A^* \Omega_g, e^{itH(g)} B \Omega_g \rangle = G_g(t).$$

Moreover  $\int_{\mathbb{R}} f G_g dt = \int_{[0,\infty)} \check{f} d\rho_g = 0$  for the stated class of  $f$  (Fubini;  $\check{f} = 0$  on the support).

*Invariance.* For  $A \in \mathfrak{A}(O)$  and fixed  $t$ , take  $n$  so large that  $d(g_n, O) > |t|$ . Then (P) and  $e^{-itH(g_n)}\Omega_{g_n} = \Omega_{g_n}$  give directly

$$\omega_{g_n}(\alpha_t(A)) = \langle \Omega_{g_n}, e^{itH(g_n)} A e^{-itH(g_n)} \Omega_{g_n} \rangle = \omega_{g_n}(A).$$

Both  $\alpha_t(A)$  and  $A$  belong to  $\mathfrak{A}(O_{|t|})$ , so norm convergence on  $\mathfrak{A}(O_{|t|})$  passes to the limit:  $\omega(\alpha_t(A)) = \omega(A)$ , and by density on  $\mathfrak{A}$ .

*Continuity and vanishing.* Fix  $A, B \in \mathfrak{A}(O)$  and  $T > 0$ . For  $n$  with  $d(g_n, O) > T$  and  $|t| \leq T$ :  $A\alpha_t(B) \in \mathfrak{A}(O_T)$  with norm  $\leq M_0$ , so

$$\sup_{|t| \leq T} |\omega(A\alpha_t(B)) - \omega_{g_n}(A\alpha_t(B))| \leq M_0 \|(\omega - \omega_{g_n})|_{\mathfrak{A}(O_T)}\| \longrightarrow 0 :$$

$t \mapsto \omega(A\alpha_t(B))$  is a locally uniform limit of the continuous functions  $G_{g_n}$ , hence continuous, bounded by  $M_0$ . For  $f$  as stated and  $\varepsilon(T) := \int_{|t| > T} |f|$ :

$$\begin{aligned} \int_{\mathbb{R}} f \omega(A\alpha_t(B)) &= \underbrace{\int_{|t| \leq T} f [\omega - \omega_{g_n}](A\alpha_t(B))}_{\leq \|f\|_1 M_0 \|(\omega - \omega_{g_n})|_{\mathfrak{A}(O_T)}\|} + \underbrace{\int_{|t| \leq T} f G_{g_n}}_{= - \int_{|t| > T} f G_{g_n}} + \int_{|t| > T} f \omega(A\alpha_t(B)), \end{aligned}$$

so its modulus is  $\leq \|f\|_1 M_0 \|(\omega - \omega_{g_n})|_{\mathfrak{A}(O_T)}\| + 2M_0 \varepsilon(T)$ ; let  $n \rightarrow \infty$ , then  $T \rightarrow \infty$ . Joint norm continuity in  $(A, B)$  extends the conclusion from local  $A, B$  to  $\mathfrak{A}$ .  $\square$

**Theorem 7.2** (Main Theorem). *Let  $0 \leq \lambda/m_0^2 < \varepsilon_{\mathcal{P}}$ . Then there is a state  $\omega_{\infty}$  on  $\mathfrak{A}$  such that:*

- (1) (full-net convergence with rate) *for every bounded open interval  $O$ ,  $\lim_{g \in \mathcal{G}} \|(\omega_g - \omega_{\infty})|_{\mathfrak{A}(O)}\| = 0$ , and  $\|(\omega_g - \omega_{\infty})|_{\mathfrak{A}(O)}\| \leq \Psi_{m_1}(d)$  whenever  $g \equiv 1$  on  $O_d$ , with  $\Psi_{m_1}$  decaying faster than every inverse power;*
- (2) (local normality)  $\omega_{\infty}|_{\mathfrak{A}(O)}$  *is normal for every  $O$ ;*
- (3) (ground state)  $\omega_{\infty}$  *is a ground state of  $(\mathfrak{A}, \alpha)$  in the spectral sense;*
- (4) (invariances)  $\omega_{\infty} \circ \tau_a = \omega_{\infty}$  *for all  $a \in \mathbb{R}$ ; and for even  $\mathcal{P}$ ,  $\omega_{\infty} \circ \theta = \omega_{\infty}$ ;*
- (5) (spectra and local moment)

$$\text{spec } H(g) \subseteq \{0\} \cup [m_1, \infty) \quad (g \in \mathcal{G}), \quad \dim \ker H(g) = 1,$$

*and  $\sup_g \|V(h)\Omega_g\| \leq M$  for unit-bounded  $h$  supported in length-2 intervals.*

*The only nonstandard weak-coupling estimates imported are the printed GJS Theorems 1.1.7–1.1.8 (Source Facts 3.1–3.2, with the cutoff uniformity of Remark 3.3). In addition, the proof uses the separately identified standard cutoff-construction inputs: self-adjointness and the simple positive cutoff ground state (Source Fact 2.2); FKN transfer, finite-volume stability, and free hypercontractivity (Lemma 3.5 and Proposition 6.1); model-specific domain smoothing and the common core (Propositions 2.4 and 6.1); finite propagation and cutoff patching; and translation/parity covariance (Source Fact 2.3 and Proposition 7.1). Proposition 2.1 verifies that the dimensionless weak-coupling hypothesis implies the source smallness condition.*

*Proof.* (5) is Theorems 5.4 and 4.1. These supply (G), (M) with  $\gamma = m_1$ , so Theorem 6.8 gives (1)–(2). For (3): choose  $\tilde{g}_n \in \mathcal{G}$  with  $\tilde{g}_n \equiv 1$  on  $(-n, n)$ ; then  $d(\tilde{g}_n, O) \rightarrow \infty$  for every  $O$  and, by (1),  $\omega_{\tilde{g}_n} \rightarrow \omega_\infty$  in norm on every  $\mathfrak{A}(O)$ ; Proposition 7.1(i) applies. For (4): translation is an order isomorphism of  $\mathcal{G}$  with cofinal range, so by (T) and (1),  $\omega_\infty(A) = \lim_g \omega_{\tau_a g}(A) = \lim_g \omega_g(\tau_{-a} A) = \omega_\infty(\tau_{-a} A)$  for local  $A$ , and by density on  $\mathfrak{A}$ . The  $\mathbb{Z}_2$  statement is the local norm limit of Proposition 7.1(iii).  $\square$

**Corollary 7.3** (the Glimm–Jaffe vacuum is canonical at weak coupling). *Every  $w^*$  limit point of the (averaged or unaveraged) finite-volume ground states constructed in [3, §2] coincides on each  $\mathfrak{A}(O)$  with  $\omega_\infty$ ; in particular the construction of [3] is subsequence- and averaging-independent in the weak-coupling region.*

*Proof.* For the unaveraged net this is Theorem 7.2(1). For the averaged states of [3, (2.3)–(2.5)]: with  $g_n = g(\cdot/n)$ ,  $g \equiv 1$  on  $[-3, 3]$ , and  $h \geq 0$ ,  $\text{supp } h \subseteq [-1, 1]$ ,  $\int h = 1$ ,

$$\begin{aligned} \omega_n(A) &= \frac{1}{n} \int \langle \Omega_{g_n}, \sigma_\alpha(A) \Omega_{g_n} \rangle h(\alpha/n) d\alpha \\ &= \int_{-1}^1 h(u) \omega_{g_n}(\tau_{\epsilon n u} A) du \\ &= \int_{-1}^1 h(u) \omega_{\tau_{-\epsilon n u} g_n}(A) du, \end{aligned}$$

by (T), where  $\epsilon \in \{\pm 1\}$  fixes the direction convention of  $\sigma_\alpha$  (the estimate below is invariant under  $\epsilon \rightarrow -\epsilon$ ). Since  $g_n \equiv 1$  on  $[-3n, 3n]$ , the shifted cutoff  $\tau_{-\epsilon n u} g_n$  equals 1 on an interval containing  $[-2n, 2n]$  for every  $|u| \leq 1$ ; hence for  $O \subseteq [-R, R]$ ,  $d(\tau_{-\epsilon n u} g_n, O) \geq 2n - R$  uniformly in  $u$ , and Theorem 7.2(1) gives

$$\|(\omega_n - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \int_{-1}^1 h(u) \|(\omega_{\tau_{-\epsilon n u} g_n} - \omega_\infty)|_{\mathfrak{A}(O)}\| du \leq \Psi_{m_1}(2n - R) \rightarrow 0.$$

No translation invariance of  $\omega_\infty$  is needed for this bound.  $\square$

**Corollary 7.4** (conditional exclusion of a two-phase structure). *Assume that  $\mathcal{P}$  is even and  $0 \leq \lambda/m_0^2 < \varepsilon_{\mathcal{P}}$ . The following four assertions cannot hold simultaneously:*

(B1) *there are translation-invariant ground states  $\omega_+$  and  $\omega_-$  of  $(\mathfrak{A}, \alpha)$  such that, for all bounded local  $X, Y$ ,*

$$\omega_\pm(X\tau_a(Y)) - \omega_\pm(X)\omega_\pm(\tau_a(Y)) \rightarrow 0 \quad (|a| \rightarrow \infty);$$

(B2)  $\omega_+ \circ \theta = \omega_-$  and  $\omega_- \circ \theta = \omega_+$ ;

(B3) *there are a bounded local observable  $B = B^* \in \mathfrak{A}(O)$  and  $m \neq 0$  such that  $\omega_\pm(B) = \pm m$ ;*

(B4) *every  $\theta$ -invariant ground state of  $(\mathfrak{A}, \alpha)$  equals  $\frac{1}{2}(\omega_+ + \omega_-)$ .*

*Proof.* Assume (B1)–(B4). Choose  $\tilde{g}_n \in \mathcal{G}$  with  $\tilde{g}_n \equiv 1$  on  $(-n, n)$  and write  $\omega_n = \omega_{\tilde{g}_n}$ . Fix  $a \in \mathbb{R}$  such that

$$R_a := \text{dist}(O, O + a) > 0, \quad m_1 R_a \geq 1.$$

For the cutoff Hamiltonian  $H(\tilde{g}_n)$  apply Lemma 6.4 with unbounded-argument slot  $X = B$ , bounded-argument slot  $A = \tau_a(B)$ , localization set  $\overline{O}$ , and  $\gamma = m_1$ . The bounded operator  $B$  has domain  $\mathcal{H}$ ; because  $B \in \mathfrak{A}(O) \subseteq \mathfrak{A}(U)$  for every bounded open  $U \supseteq \overline{O}$ , it is affiliated

with every such  $\mathfrak{A}(U)$ . Thus all hypotheses of that lemma hold. Since  $\|B\Omega_{\tilde{g}_n}\| \leq \|B\|$  and  $\|\tau_a(B)\| = \|B\|$ , it gives, uniformly in  $n$ ,

$$|\omega_n(B\tau_a(B)) - \omega_n(B)\omega_n(\tau_a(B))| \leq 2\|B\|^2 e^{-m_1 R_a/2}. \quad (17)$$

For fixed  $a$ , all three observables in (17) belong to the local algebra of one bounded interval containing  $O \cup (O + a)$ . The local-norm convergence of Theorem 7.2(1) therefore permits passage to  $n \rightarrow \infty$ . Translation invariance of  $\omega_\infty$  then yields

$$|\omega_\infty(B\tau_a(B)) - \omega_\infty(B)^2| \leq 2\|B\|^2 e^{-m_1 R_a/2} \xrightarrow{|a| \rightarrow \infty} 0.$$

By Theorem 7.2(3)–(4),  $\omega_\infty$  is a  $\theta$ -invariant ground state. Assumption (B4) consequently gives

$$\omega_\infty = \frac{1}{2}(\omega_+ + \omega_-), \quad \omega_\infty(B) = \frac{1}{2}(m - m) = 0.$$

By (B1), translation invariance and clustering of each  $\omega_\pm$  imply

$$\omega_\pm(B\tau_a(B)) - \omega_\pm(B)\omega_\pm(\tau_a(B)) \longrightarrow 0, \quad \omega_\pm(\tau_a(B)) = \omega_\pm(B) = \pm m.$$

Hence

$$\omega_\infty(B\tau_a(B)) = \frac{1}{2}(\omega_+(B\tau_a(B)) + \omega_-(B\tau_a(B))) \longrightarrow \frac{1}{2}(m^2 + m^2) = m^2 \neq 0,$$

contradicting the preceding cluster bound, whose left side tends to zero because  $\omega_\infty(B) = 0$ .  $\square$

## 8 Discussion

*Relation to GRS Problems 1–2.* Problem 2 of [10, p. 126] asks for local convergence of Euclidean Gibbs states; Theorem 7.2(1) is a Hamiltonian counterpart, in norm on local von Neumann algebras, for the vacuum family. Problem 1 asks for uniform local  $L^p$  bounds on the Euclidean vacuum density. The note added on [10, p. 128] records Newman’s solution of the small-coupling  $p = 1$  versions, so the printed problems must not be read as an unqualified current open-problem claim. The present proof neither establishes nor needs Problem 1: the identification step uses the exact spectral identity of Proposition 7.1, and the convergence step uses the Cauchy estimate.

*Sharpness.* In the quadratic model the exact rate is  $\text{poly}(d) e^{-2m_1 d}$  (companion M2: round trip through the region where the Hamiltonians differ, at the local Agmon mass).  $\Psi_{m_1}$  is super-polynomial but sub-exponential. The present filter argument does not obtain the sharp exponential rate; a proof of such a rate would require an additional resolvent or Combes–Thomas input not supplied here.

*Uniqueness.*  $\omega_\infty$  is canonical for the cutoff family, and translation invariant without averaging. The theorem does not identify every locally normal translation-invariant ground state of  $(\mathfrak{A}, \alpha)$  with  $\omega_\infty$ . In particular, the Euclidean results [13, 14] do not by themselves provide the reverse reconstruction and state-space identification needed for that Hamiltonian conclusion; companion M3 records that missing implication as (R3).

*Beyond weak coupling.* Corollary 7.4 is conditional on (B1)–(B4). Under exactly those hypotheses, a proof based on a cutoff-uniform gap cannot extend to that two-phase regime. The phase-transition results [8, 9], as cited here, are not used as a substitute for (B4), and this manuscript makes no unconditional assertion that its Hamiltonian state space has the classification (B1)–(B4).

*Part II.* Uniformity in a wavelet resolution — the OAR programme — is a separate problem. For the MMST free nearest-neighbour model and its Daubechies observable embedding [16], the companion Lamport proof constructs local Weyl unitaries for which

$$\liminf_{N \rightarrow \infty} \|\omega_N - \omega_\infty \circ \beta_N\|_{\mathfrak{W}_N(O)} \geq e^{-\sqrt{2+\sqrt{2}}/4} - e^{-3\pi/16} > 0.$$

Thus no diagonal local-state-norm error  $C\varepsilon_N^\theta$ ,  $\theta > 0$ , is possible. The implemented replacement fixes a torus  $\mathbb{T}_L$ , the MMST nearest-neighbour Hamiltonian, the exact  $D4$  observable maps, and the finite-torus Wick convention. The finite-resolution interacting Schrödinger form is closed and lower bounded; its operator has compact resolvent and a simple ground state.

For these interacting ground states, weak-star compactness gives a cofinal subnet  $N(i)$  and a state  $\Omega_{L,\lambda}^{\text{cl}}$  on the inductive-limit algebra such that, for every fixed  $M$  and  $A \in \mathfrak{W}_{M,L}$ ,

$$\lim_i \omega_{N(i),L,\lambda,1}(\alpha_M^{N(i)}(A)) = \Omega_{L,\lambda}^{\text{cl}}(\alpha_M^\infty(A)).$$

This proves existence of a projective interacting cluster state, not convergence of the full sequence or its identification with the continuum vacuum. In the free case the full sequence satisfies the explicit estimate

$$\begin{aligned} & \left| \omega_{M+r,L}^{(0)} \circ \alpha_M^{M+r}(W_M(\xi)) - \omega_L^{(0)} \circ \beta_M(W_M(\xi)) \right| \\ & \leq \frac{C_{M,L,m,\xi}}{4} 2^{-\theta r}, \quad \theta = \frac{4(1 - \log_2 \sqrt{3})}{7 - 2 \log_2 \sqrt{3}} > 0. \end{aligned}$$

For the static interacting theorem, the exact remaining assertion is

$$\lim_{N \rightarrow \infty} \omega_{N,L,\lambda,1}(\alpha_M^N(W_M(\xi))) = \omega_{P(\phi)_2,L}(\beta_M(W_M(\xi))) \quad \text{for every fixed } M, \xi. \quad (\text{PII})$$

The companion Weyl criterion proves that (PII), by itself, extends to every  $A \in \mathfrak{W}_{M,L}$ , gives projective consistency, and identifies the inductive-limit state. A uniform gap, local second moment, and interacting Lieb–Robinson bound are therefore not prerequisites for this static fixed-torus criterion. They remain necessary for stronger quantitative-locality or dynamical claims. In particular, Theorem 4.1 of [17] assumes  $V' \in L^1(\mathbb{R})$  and cannot be applied to the Wick quartic, since  $|V'_c(u)| \geq 2\lambda|u|^3$  for  $|u| \geq \sqrt{6c}$ . No full-sequence identified interacting resolution-limit theorem is asserted here. The proofs and exact ledgers are in `strategy.md` and the five Lamport documents in `part_ii/`.

## A Interface checklist

- (G), (M) of `lpp1.tex`: supplied with  $\gamma = m_1$ ,  $M$  as in Theorem 4.1. Hypothesis classes match  $(\mathcal{G}; \|h\|_\infty \leq 1, \text{ support in length-2 intervals})$ . ✓
- (P), (T): supplied by Source Fact 2.3. ✓
- Simplicity of  $E(g)$  (used in Lemma 3.6 and Prop. 7.1(iii)): Source Fact 2.2. No isolation input anywhere. ✓
- $\Omega_g \in D(V(h))$  (used in Prop. 6.4): Prop. 2.4(i), and independently re-proved in Theorem 4.1 Step 3. ✓
- Wick/covariance conventions between [7] and §2: Remark 3.4. ✓
- Gate R2: *closed by page-image inspection* (`m4/r2-check.tex`) —  $K_1$  to the first power, local weight  $K_2^{2N(\Delta)}$ ; all displayed constants use the exact reading. ✓

## References

- [1] J. Glimm, A. Jaffe, Phys. Rev. **176** (1968) 1945–1951.
- [2] J. Glimm, A. Jaffe, Ann. of Math. **91** (1970) 362–401.
- [3] J. Glimm, A. Jaffe, Acta Math. **125** (1970) 203–267.
- [4] J. Glimm, A. Jaffe, J. Math. Phys. **13** (1972) 1568–1584.
- [5] J. Glimm, A. Jaffe, Comm. Math. Phys. **22** (1971) 253–258.
- [6] J. Glimm, A. Jaffe, *Quantum Field Theory and Statistical Mechanics: Expositions*, Birkhäuser, 1985.
- [7] J. Glimm, A. Jaffe, T. Spencer, Ann. of Math. **100** (1974) 585–632.
- [8] J. Glimm, A. Jaffe, T. Spencer, Comm. Math. Phys. **45** (1975) 203–216.
- [9] J. Glimm, A. Jaffe, T. Spencer, Ann. Physics **101** (1976) 610–669.
- [10] F. Guerra, L. Rosen, B. Simon, Ann. of Math. **101** (1975) 111–259.
- [11] L. Rosen, Comm. Math. Phys. **16** (1970) 157–183.
- [12] B. Simon, *The  $P(\Phi)_2$  Euclidean (Quantum) Field Theory*, Princeton UP, 1974.
- [13] S. Albeverio, R. Høegh-Krohn, B. Zegarlinski, Comm. Math. Phys. **123** (1989) 377–424.
- [14] R. Bauerschmidt, B. Dagallier, H. Weber, arXiv:2504.08606.
- [15] M. Rodriguez Zarate, T. Thiemann, arXiv:2505.13030.
- [16] V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, Commun. Math. Phys. **387** (2021) 1299–1367.
- [17] B. Nachtergaele, H. Raz, B. Schlein, R. Sims, Commun. Math. Phys. **286** (2009) 1073–1098.